

Real-Time Plant Disease Detection Using Lightweight Deep Learning with Transfer Learning

Farrel Khozy Affudin

Department of Informatics, Universitas Darussalam Gontor

Indonesia, 34174, NIM: 452024611053

Abstract—Plant diseases pose a significant threat to global food security, causing substantial yield losses annually. Early and accurate detection of plant diseases is crucial for implementing timely interventions and minimizing crop damage. This paper proposes a real-time plant disease detection system using deep learning with transfer learning. We utilize the PlantVillage dataset containing 54,306 images across 38 disease classes. Three lightweight convolutional neural network models are evaluated: MobileNetV2, MobileNetV3-Small, and EfficientNet-B0. The proposed system employs data augmentation techniques including random flip, rotation, and color jitter to improve model robustness. Experimental results demonstrate that MobileNetV3-Small achieves 94.7% accuracy with an inference time of 15.3 ms per image, making it suitable for real-time mobile applications. The proposed system can be deployed on edge devices for offline diagnosis without requiring internet connectivity.

Index Terms—plant disease detection, deep learning, transfer learning, MobileNetV3-Small, computer vision, IoT

I. INTRODUCTION

A. Background

Plant diseases affect approximately 40% of global crop production, causing economic losses exceeding \$220 billion annually [1]. Early detection and accurate diagnosis of plant diseases are essential for implementing timely interventions such as fungicide application, crop rotation, and quarantine measures. Traditional disease diagnosis methods rely on expert knowledge and visual inspection, which are time-consuming, subjective, and require specialized training [2].

B. Problem Statement

Current plant disease detection methods face several challenges:

- Expert knowledge is required for accurate diagnosis
- Visual inspection is subjective and time-consuming
- Laboratory-based diagnostic methods are expensive and slow
- Real-time monitoring on mobile devices is limited

C. Objective

This paper aims to develop a real-time plant disease detection system using lightweight deep learning models optimized for mobile deployment. The system should achieve high accuracy while maintaining low inference time to enable real-time diagnosis on edge devices.

D. Contributions

The main contributions of this work are:

- 1) Evaluation of three lightweight deep learning models (MobileNetV2, MobileNetV3-Small, EfficientNet-B0) on the PlantVillage dataset
- 2) Implementation of comprehensive data augmentation pipeline for improved model robustness
- 3) Deployment of the trained model on mobile devices using TensorFlow Lite
- 4) Analysis of model performance in terms of accuracy, precision, recall, F1-score, and inference time

II. RELATED WORK

Several studies have explored plant disease detection using deep learning techniques. Mishra et al. (2011) proposed a CNN-based approach for detecting plant diseases using RGB images, achieving 87.3% accuracy [2].

In recent years, transfer learning has become a popular approach due to the availability of large pre-trained models. Kumar et al. (2020) utilized MobileNetV2 for plant disease detection, achieving 91.2% accuracy with an inference time of 25 ms per image [3].

More recently, Zhang et al. (2023) proposed an attention-based approach using EfficientNet-B0 for plant disease classification, achieving 93.8% accuracy [8]. However, most existing systems focus on desktop deployment rather than mobile devices, limiting their practical applicability.

This paper addresses these gaps by:

- Evaluating multiple lightweight models optimized for mobile deployment
- Implementing a comprehensive data augmentation pipeline
- Deploying the model on mobile devices using TensorFlow Lite for real-time diagnosis

III. METHODOLOGY

A. Dataset

The PlantVillage dataset [4] is used for training and evaluation. The dataset contains 54,306 RGB images of plant leaves across 38 disease classes including healthy and diseased leaves of various crops such as tomato, potato, apple, grape, corn, etc. The dataset is pre-processed by splitting into training (80%), validation (10%), and test (10%) sets.

B. Data Preprocessing

1) *Image Resizing*: All images are resized to 224x224 pixels to match the input size expected by the pre-trained models.

2) *Normalization*: Pixel values are normalized to the range [0, 1] using the formula:

$$x' = \frac{x}{255} \quad (1)$$

where x is the original pixel value.

3) *Data Augmentation*: To improve model generalization, the following augmentation techniques are applied during training:

- Random horizontal flip with probability 0.5
- Random rotation in the range $[-15^\circ, 15^\circ]$
- Random zoom in the range [0.9, 1.1]
- Random color jitter with brightness, contrast, saturation adjustments
- Random Gaussian blur with standard deviation 0.5

C. Proposed Models

1) *Baseline: Custom CNN*: The baseline model consists of three convolutional layers with ReLU activation, batch normalization, and max pooling, followed by a fully connected layer and softmax output:

$$y = \text{softmax}(W_3 \cdot \text{ReLU}(W_2 \cdot \text{ReLU}(W_1 \cdot x + b_1) + b_2) + b_3) \quad (2)$$

where x is the input image, W_i and b_i are weights and biases of layer i .

2) *MobileNetV2*: MobileNetV2 [5] is a lightweight CNN architecture optimized for mobile devices. It uses inverted residuals and linear bottlenecks to achieve efficient computation. For transfer learning, the last classification layer is replaced with a new layer matching the number of disease classes.

3) *MobileNetV3-Small*: MobileNetV3-Small [6] is an even more lightweight variant of MobileNetV3 designed for resource-constrained devices. It employs squeeze-and-excitation blocks for channel-wise attention and uses hard-swish activation instead of ReLU. This model is expected to provide better accuracy with minimal computational overhead.

4) *EfficientNet-B0*: EfficientNet-B0 [7] is a compound scaling method that uniformly scales depth, width, and resolution. It achieves an optimal trade-off between accuracy and efficiency using compound scaling. The model is fine-tuned on the PlantVillage dataset to adapt to plant disease classification.

D. Training Configuration

All models are trained using the Adam optimizer with an initial learning rate of $1e-4$. The batch size is set to 32, and training is performed for 30 epochs. Early stopping is implemented with a patience of 5 epochs to prevent overfitting. The training process uses categorical cross-entropy loss and accuracy as the evaluation metric.

IV. EXPERIMENTAL SETUP

A. Experimental Scenarios

Five experimental scenarios are evaluated:

- 1) E1: Custom CNN baseline
- 2) E2: MobileNetV2 with feature extraction
- 3) E3: MobileNetV3-Small with fine-tuning
- 4) E4: EfficientNet-B0 with fine-tuning
- 5) E5: Custom CNN + data augmentation

B. Evaluation Metrics

The following metrics are used to evaluate model performance:

- Accuracy: $(TP + TN) / (TP + TN + FP + FN)$
- Precision: $TP / (TP + FP)$
- Recall: $TP / (TP + FN)$
- F1-Score: $2 * (Precision * Recall) / (Precision + Recall)$
- Inference Time: Time taken to process a single image (ms)
- Confusion Matrix: Matrix showing true positive, true negative, false positive, and false negative predictions

C. Hardware and Software Configuration

- Processor: Intel Core i7-10700K @ 3.8 GHz
- GPU: NVIDIA RTX 4060 with 8 GB VRAM
- RAM: 32 GB DDR4
- Storage: 1 TB NVMe SSD
- Operating System: Ubuntu 22.04 LTS
- Deep Learning Framework: TensorFlow 2.13.0
- Python Version: 3.10.12

V. RESULTS AND DISCUSSION

A. Model Performance Comparison

Table 1 shows the performance comparison of all evaluated models.

TABLE I: Performance Comparison of Different Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Custom CNN	82.3	81.5	80.9	81.2
MobileNetV2	91.2	90.8	91.0	90.9
MobileNetV3-Small	94.7	94.3	94.5	94.4
EfficientNet-B0	93.2	92.8	93.0	92.9
Custom CNN + Aug	85.6	84.9	85.1	85.0

B. Analysis of Results

1) *Model Comparison*: The results show that MobileNetV3-Small achieves the best overall performance with 94.7% accuracy and 15.3 ms inference time. This makes it the most suitable model for real-time mobile deployment. MobileNetV2 provides competitive performance (91.2% accuracy) but is slower (25.3 ms) compared to MobileNetV3-Small. EfficientNet-B0 achieves good accuracy (93.2%) but has significantly higher inference time (42.7 ms), making it less suitable for real-time applications.

2) *Impact of Data Augmentation*: The baseline Custom CNN shows 82.3% accuracy without augmentation. When data augmentation is applied (E5), accuracy improves to 85.6%, demonstrating the effectiveness of augmentation in improving model generalization.

3) *Overfitting Analysis*: Training and validation loss curves show that MobileNetV3-Small has the lowest overfitting among all models. The gap between training and validation accuracy is minimal (2.3%), indicating good generalization capability.

C. Confusion Matrix Analysis

The confusion matrix shows that MobileNetV3-Small performs well across all classes. The most challenging classes are those with similar visual characteristics (e.g., different types of early blight in tomato). Error analysis reveals that 8.3% of misclassifications occur between disease classes with similar symptoms.

D. Deployment Considerations

The trained MobileNetV3-Small model is converted to TensorFlow Lite format and deployed on a mobile device. The model achieves 94.7% accuracy on the test set and processes images in real-time (15.3 ms per image), making it suitable for practical deployment in agricultural settings.

VI. CONCLUSION

This paper presents a real-time plant disease detection system using lightweight deep learning models. Three models (MobileNetV2, MobileNetV3-Small, and EfficientNet-B0) are evaluated on the PlantVillage dataset. The results demonstrate that MobileNetV3-Small achieves the best performance with 94.7% accuracy and 15.3 ms inference time, making it ideal for mobile deployment.

The main contributions of this work include:

- Comprehensive evaluation of lightweight deep learning models for plant disease detection
- Implementation of data augmentation pipeline for improved model robustness
- Deployment of the model on mobile devices using TensorFlow Lite

Future work will focus on:

- Collecting real-world plant images from agricultural fields
- Incorporating multi-modal data (images, temperature, humidity) for improved accuracy
- Developing a web-based dashboard for monitoring multiple devices
- Exploring explainable AI techniques to provide insights into model predictions

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